

Assignment 3: End-to-End Hugging Face Model Training and Docker Deployment

1 Introduction

This assignment demonstrates a complete end-to-end machine learning workflow using the Hugging Face ecosystem. The objective was to convert a research notebook into production-ready scripts, train and evaluate a transformer-based model, containerize the workflow using Docker, and publish the model artifacts.

The task involved dataset preparation, model training using the Hugging Face Trainer API, evaluation, containerization, and repository publication.

2 Model Selection

DistilBERT (`distilbert-base-cased`) was selected for this task.

Reason for Selection:

- DistilBERT is a lightweight and distilled version of BERT
- Faster training and inference compared to full BERT
- Reduced memory requirements
- Suitable for containerized environments
- Maintains strong performance on text classification tasks

These characteristics make DistilBERT well suited for resource-constrained environments such as Docker containers.

3 Dataset

The model was trained on Goodreads review data categorized by book genre. The dataset was sampled to create balanced training and test splits.

- Number of classes: 8 genres
- Training samples: 6,400
- Test samples: 1,600

4 Training Summary

The model was fine-tuned using the Hugging Face Trainer API with the following configuration:

- Pre-trained model: DistilBERT base cased
- Epochs: 3
- Learning rate: 5×10^{-5}
- Batch size (train): 10
- Batch size (eval): 16
- Warmup steps: 100
- Weight decay: 0.01

Training was performed inside a Docker container.

5 Evaluation Results

Model performance was assessed on the held-out test dataset using accuracy, precision, recall, and F1-score.

5.1 Classification Report

Genre	Precision	Recall	F1-score	Support
Children	0.64	0.66	0.65	200
Comics & Graphic	0.79	0.83	0.81	200
Fantasy & Paranormal	0.46	0.49	0.47	200
History & Biography	0.65	0.52	0.58	200
Mystery/Thriller/Crime	0.55	0.58	0.57	200
Poetry	0.80	0.78	0.79	200
Romance	0.62	0.63	0.63	200
Young Adult	0.42	0.41	0.41	200
Accuracy	0.61 (1600 samples)			
Macro Average	0.62	0.61	0.61	1600
Weighted Average	0.62	0.61	0.61	1600

5.2 Performance Analysis

The model achieved moderate performance across most genres, with the best results observed for:

- Comics & Graphic
- Poetry

Lower performance was observed for:

- Young Adult
- Fantasy & Paranormal

This likely reflects overlapping themes between these genres, making classification more difficult.

6 Model Deployment to Hugging Face

The trained model, tokenizer, and configuration files were uploaded to the Hugging Face Hub and made publicly accessible.

Repository Link:

<https://huggingface.co/Apollo112/bert-goodreads-genre-assignment3/tree/main>

The model can be downloaded and used directly via the `from_pretrained()` API.

7 Docker Implementation

The entire workflow was containerized using Docker to ensure reproducibility and environment consistency. Separate images were used for training and evaluation phases.

8 Challenges Faced

Several practical issues were encountered during the implementation:

- Multiple debugging cycles were required while converting notebook code into modular scripts.
- Docker containers initially failed to run due to insufficient file system write permissions.
- Training time was very long (approximately 40 minutes to 1 hour per run), and errors sometimes appeared only after training completed, requiring repeated executions.
- Python module mismatches caused compatibility issues. In particular, the code required PyTorch 2.4.0, while the system initially had PyTorch 2.2.0 installed.
- Additional challenges included resolving dependency conflicts, modifying Dockerfiles multiple times, and addressing permission-related issues on the host system.
- Splitting the notebook into modular Python files while maintaining correct data flow between modules introduced further debugging complexity.